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Demonstrating real advantage of machine-learning-enhanced Monte Carlo for combinatorial optimization

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PNAS (2026)

Motivations

Goal: generate equilibrium samples from a Boltzmann distribution $P_B(\sigma) = \dfrac{e^{-\beta H(\sigma)}}{Z}$

For $\beta \to \infty$ find the minimum of $H(\sigma)$ - combinatorial optimization

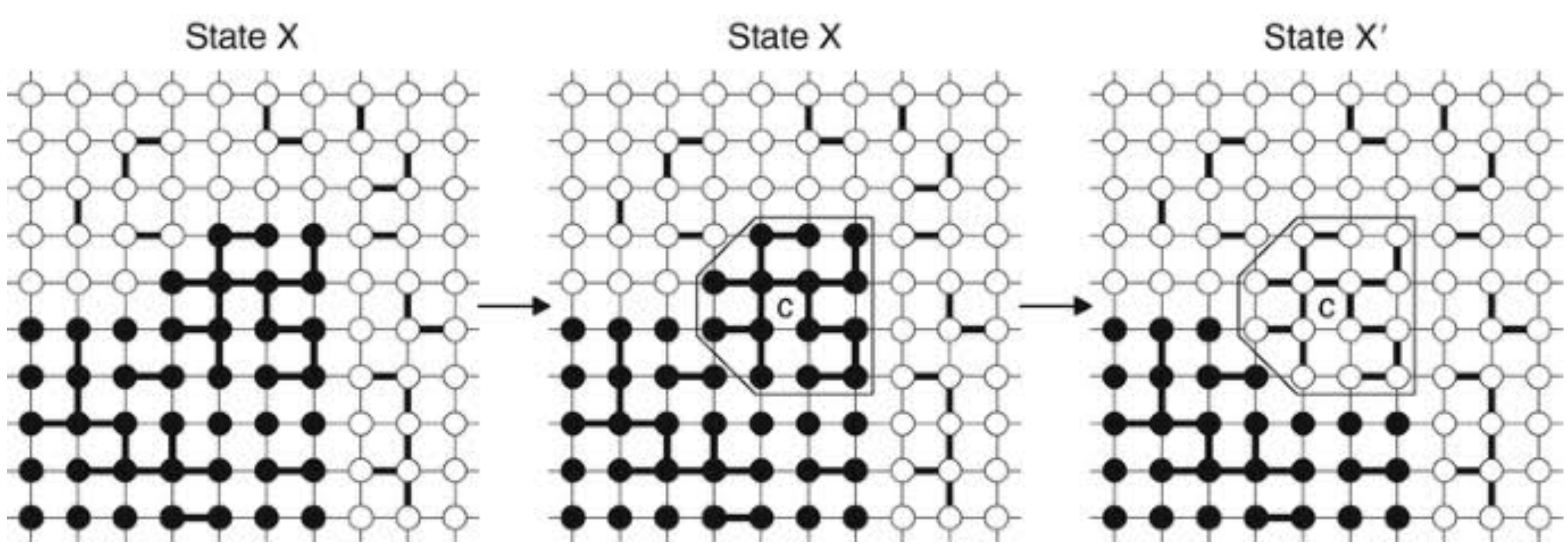
Universal strategy: local MCMC + parallel tempering + population annealing + ...

But: there are hard-to-sample problems, among which glasses, NP-hard optimization…

Theorem: slow local decorrelation is due to spatial correlations between variables

Montanari, Semerjian (2006)

Propose 'smart moves' to update clusters of correlated variables?



A universal solution?

Global MCMC scheme - 'learn' a model that generates the right moves

1. Propose a new configuration σ_{new} sampled from a generative model $P_A(\sigma)$

2. Accept the move with Metropolis-Hastings rate

$$\mathrm{Acc}\left[\sigma_{old} \to \sigma_{new}\right] = \min\left[1, \frac{P_B(\sigma_{new}) \times P_A(\sigma_{old})}{P_B(\sigma_{old}) \times P_A(\sigma_{new})}\right]$$

NB: similar to importance sampling - but can (and must!) be combined with local MCMC

Gabrié, Rotskoff, Vanden-Ejiden (2021)
Del Bono, Ricci-Tersenghi, FZ (PRE 2025)

It will work if:

1. The reweighing $W(\sigma) = P_B(\sigma)/P_A(\sigma)$ does not fluctuate wildly

2. States from $P_A(\sigma)$ cover well the support of $P_B(\sigma)$

$\Bigg\} \Leftrightarrow D_{KL}(P_B || P_A)$ is small

$$D_{KL}(P_B || P_A) = \left\langle \log W(\sigma) \right\rangle_{P_B} \ll N$$

Learn P_A by minimizing $D_{KL}(P_B || P_A)$

The dangers of improper benchmarking

Article | Published: 21 April 2022

Combinatorial optimization with physics-inspired graph neural networks

Martin J. A. Schuetz, J. Kyle Brubaker & Helmut G. Katzgraber

Nature Machine Intelligence 4, 367–377 (2022) | Cite this article

Matters Arising to this article was published on 30 December 2022

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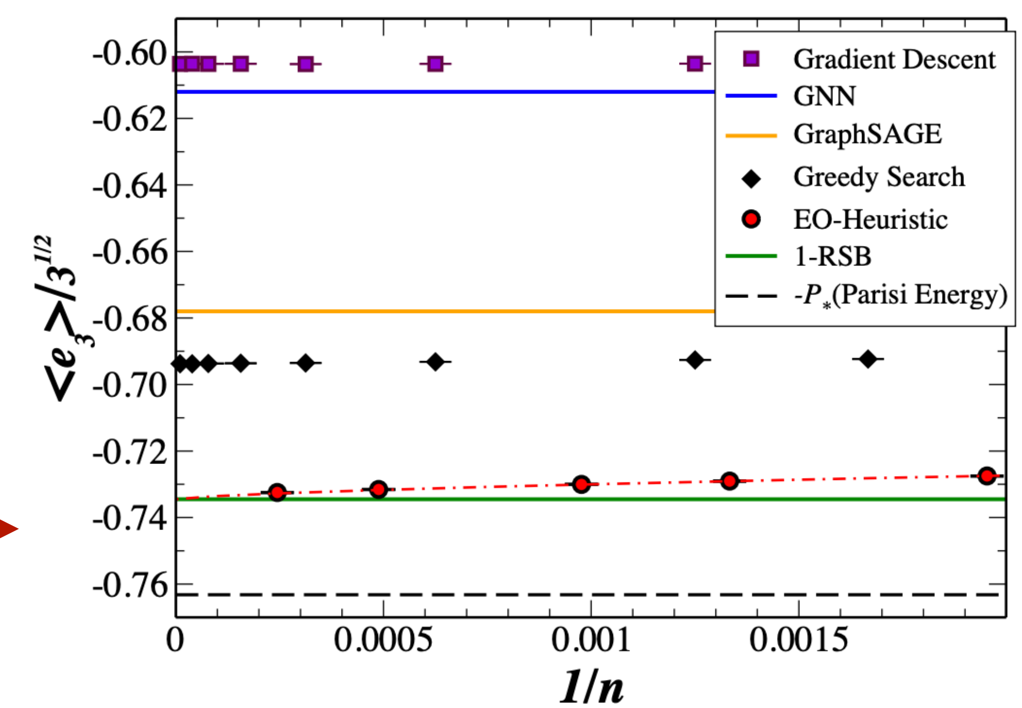
Matters Arising | Published: 30 December 2022

Inability of a graph neural network heuristic to outperform greedy algorithms in solving combinatorial optimization problems

Stefan Boettcher

Nature Machine Intelligence 5, 24–25 (2023) | Cite this article

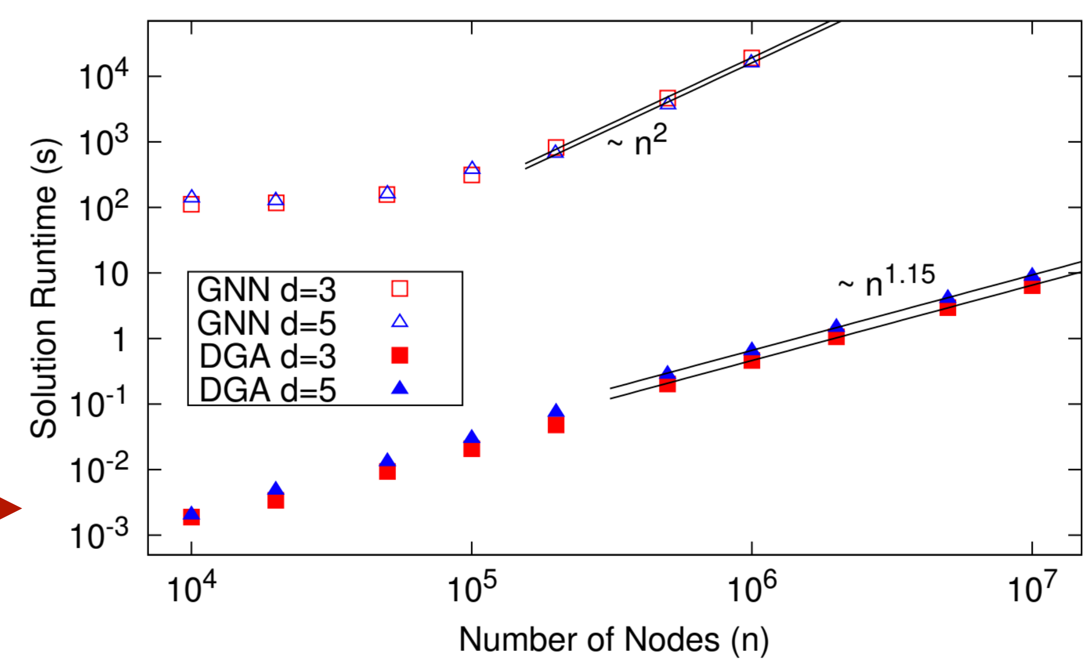


Matters Arising | Published: 30 December 2022

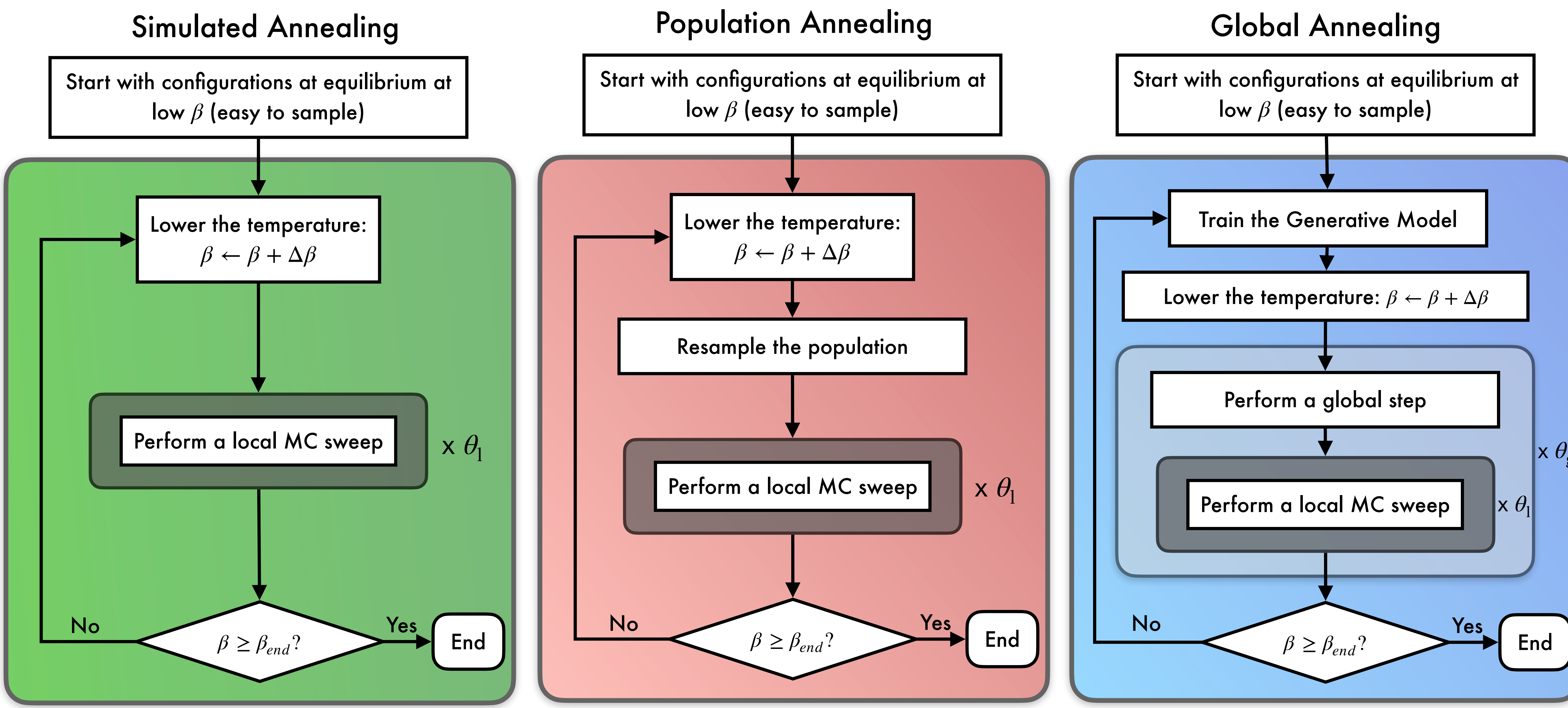
Modern graph neural networks do worse than classical greedy algorithms in solving combinatorial optimization problems like maximum independent set

Maria Chiara Angelini & Federico Ricci-Tersenghi

Nature Machine Intelligence 5, 29–31 (2023) | Cite this article

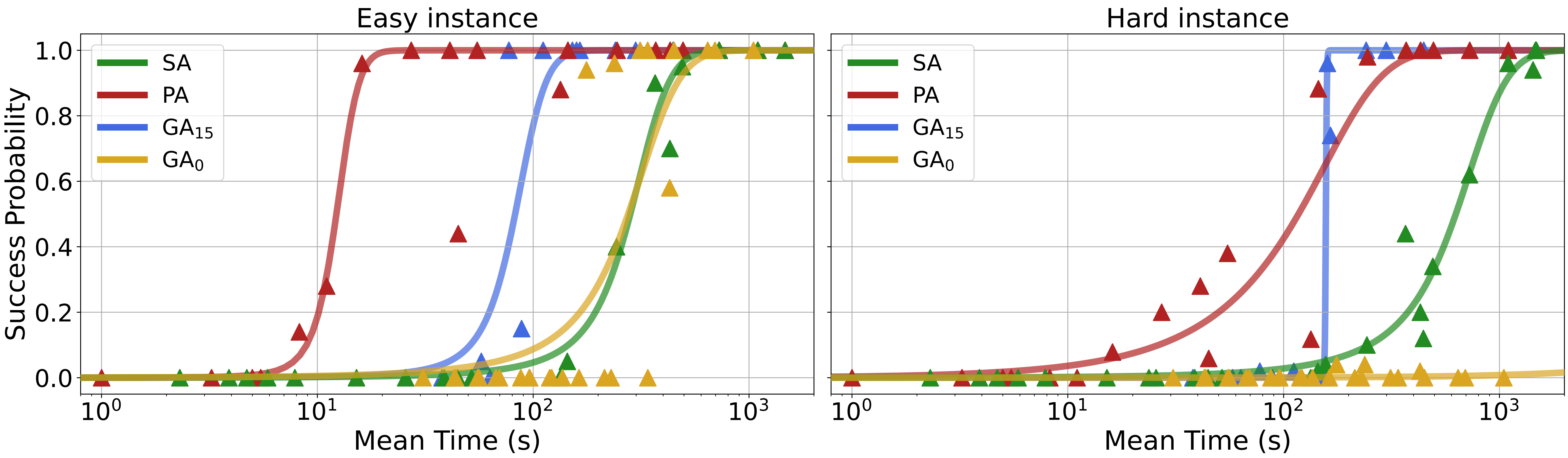


Algorithms



- Identical implementation on a single GPU (using MADE for GA)
- Validated benchmark instances of Quadratic Unconstrained Binary Optimization: random instances of 3dEA, check instance-to-instance fluctuations
- Fair wall-clock time comparison
- Check scalability with system size

Two instances of 3dEA



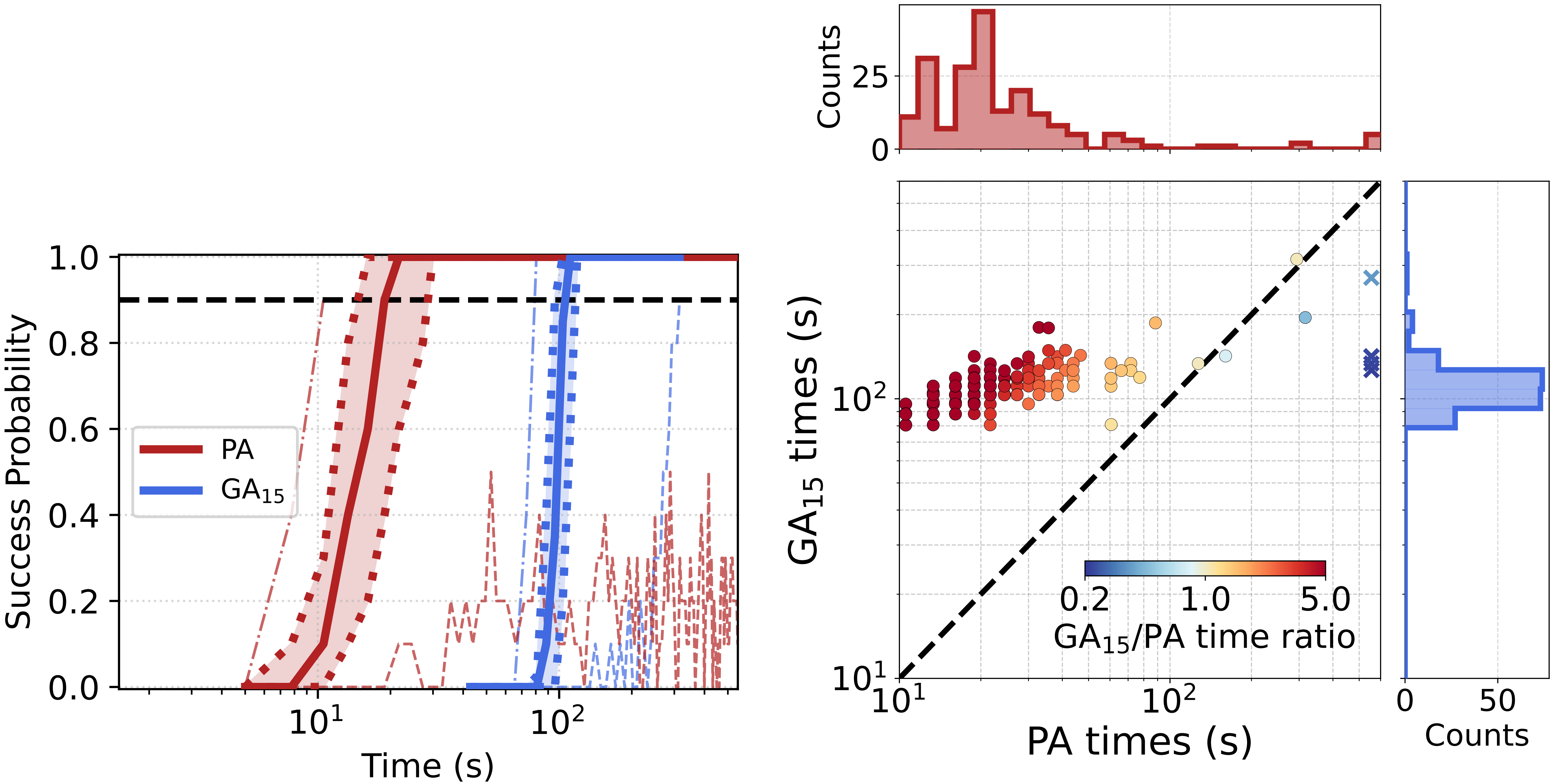
One `easy', one `hard' instance with $L = 10, N = 1000$, success over 50 runs

GA_0 does not work - need for local moves

Easy instance: PA outperforms GA

Hard instance: PA ~ GA, but GA is much sharper

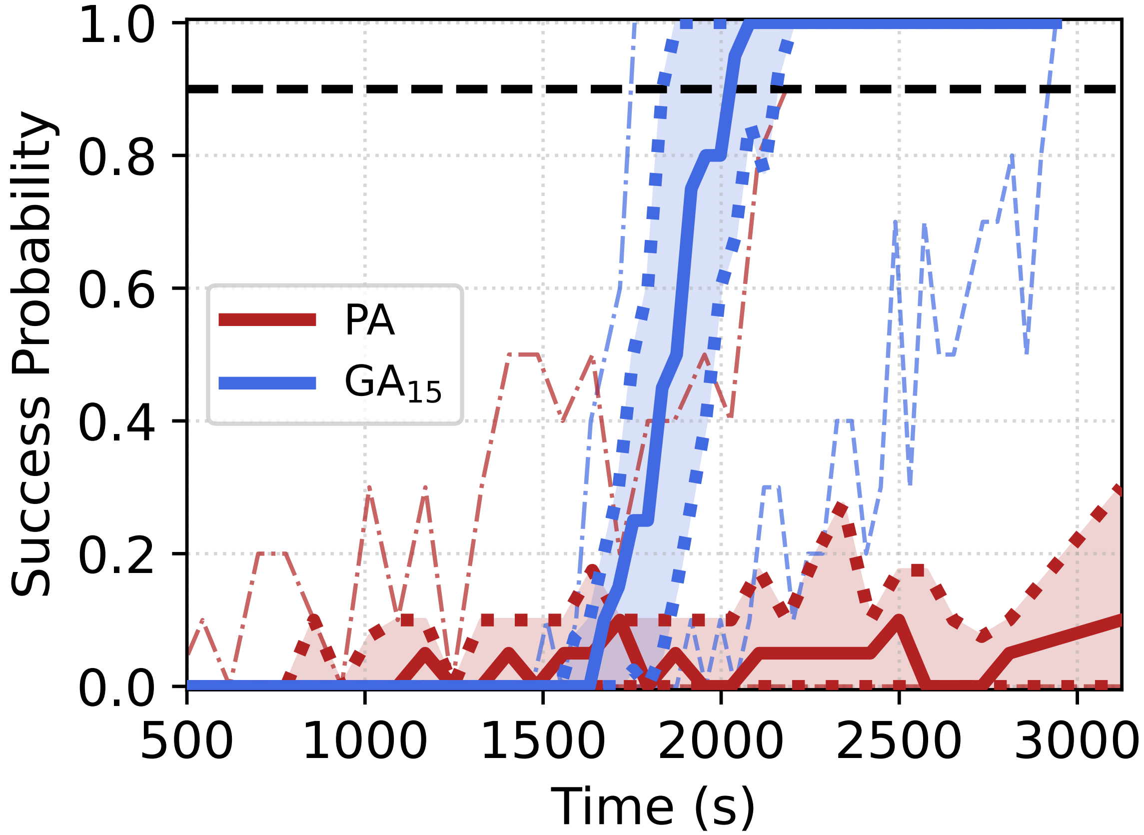
Many instances of 3dEA



200 instances with $L = 10, N = 1000$, success over 10 runs

GA is more robust to instance-to-instance fluctuations
GA is more efficient for very hard instances

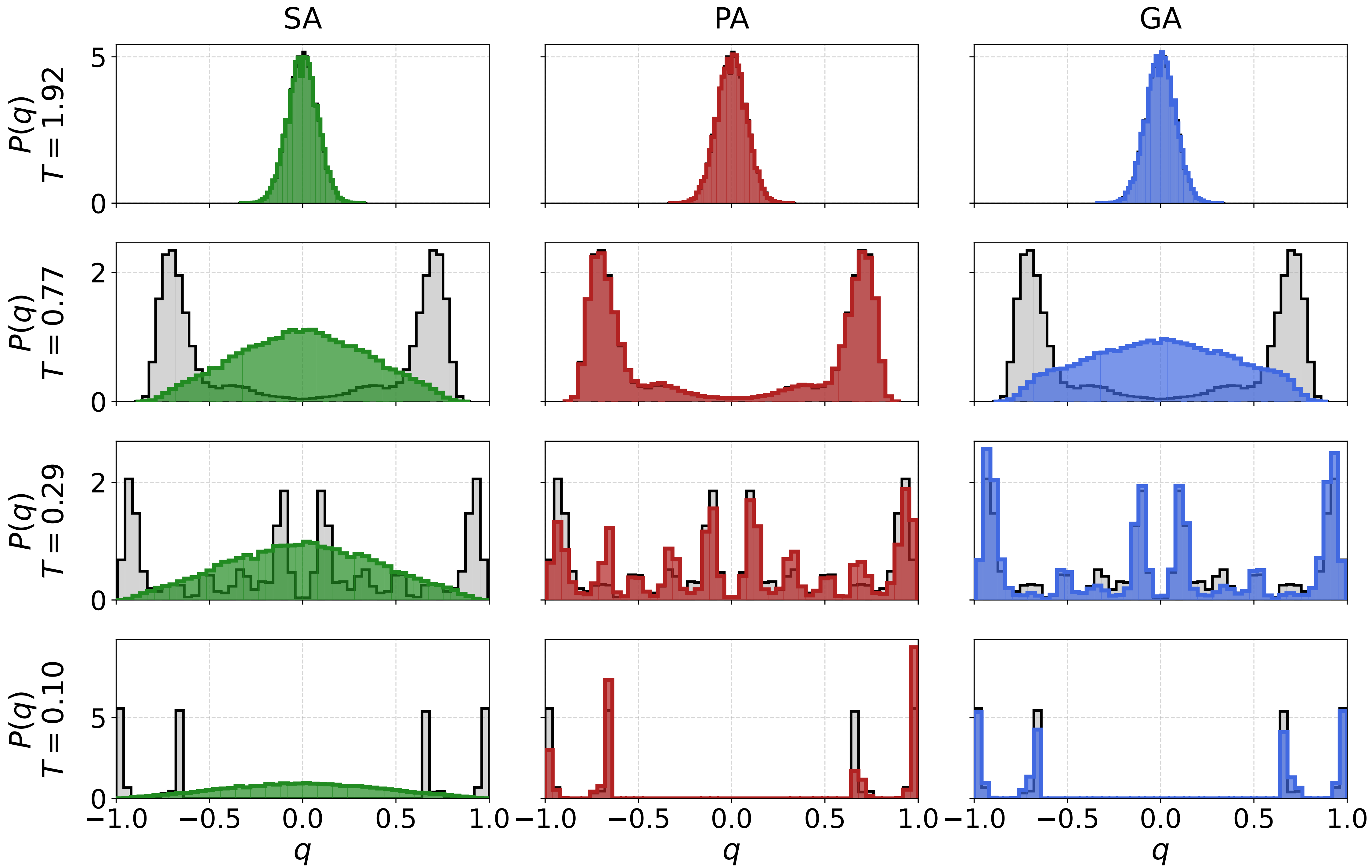
Larger instances of 3dEA



10 instances with $L = 14, N = 2744$, success over 10 runs

GA is more efficient and more robust for large instances

Overlap probability distribution



A successful run on the `hard' instance with $L = 10, N = 1000$

SA relies on rare events - one chain finds the ground state
PA relies on improved thermalization by reweighting - needs a large population
GA does the same more efficiently via the generative model

Some progress

- Architectures scale well with $O(N)$ parameters. ✔
- Not transferable: one has to learn a specific model for each sample of the problem. ?
- Global annealing is robust against instance-to-instance fluctuations ✔
- It can be used to find the ground state efficiently in 2dEA and 3dEA ✔
- What about really hard problems with exponentially many states? ✘

Ciarella, Trinquier, Weigt, FZ (2023)

Perspectives

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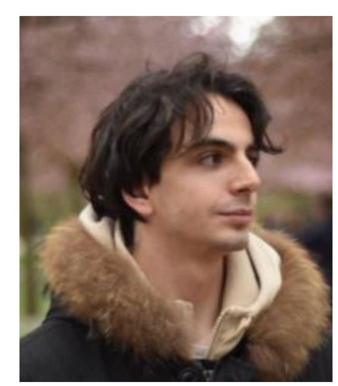
- Better understanding of the role of correlations during annealing
- Use entirely different architectures, e.g. diffusion models
- Use the couplings as input to the GNN. Transferable architectures?
- Study harder problems, e.g. lattice glass or XORSAT
- Construct good benchmarks for further testing

Thank you for your attention!



Simone Ciarella, Jeanne Trinquier, Martin Weigt
Machine Learning: Science and Technology (2023)




Luca Maria Del Bono, Federico Ricci-Tersenghi
Machine Learning: Science and Technology (2025)
Physical Review E (2025)
PNAS (2026)